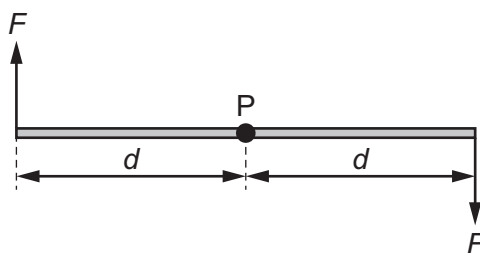


- 14 Two forces, each of magnitude F , act in opposite directions on a rod.



Each force acts on the rod at a distance d from the pivot P.

What is the torque of this couple about P?

- A** 0 **B** $F \times d$ **C** $2F \times d$ **D** $2F \times 2d$

- 15 A man of weight 600 N stands with both feet flat on the ground.

What is a reasonable estimate of the pressure exerted on the ground by the weight of the man?

- A** $1 \times 10^0 \text{ Pa}$ **B** $1 \times 10^2 \text{ Pa}$ **C** $1 \times 10^4 \text{ Pa}$ **D** $1 \times 10^6 \text{ Pa}$

- 16 The formula for the upthrust on a block of wood partially submerged in water is shown.

$$F = \rho g V$$

What do the symbols ρ and V represent?

	ρ	V
A	density of water	volume of whole block
B	density of water	volume of block below surface of water
C	density of wood	volume of whole block
D	density of wood	volume of block below surface of water

- 17 An object with a mass of 100 g falls a vertical distance of 10 m.

What is the change in the gravitational potential energy of the object?

- A** 1 J **B** 10 J **C** 100 J **D** 10 000 J